

Answer to the Question NO:01

a) $(5.2378)_{10} = (5 + 0.2378)_{10}$

Now,

$$\begin{array}{r} 2 \overline{) 5} \\ 2 \overline{) 2} \quad 1 \\ 2 \overline{) 1} \quad 0 \\ \hline 0 \quad 1 \end{array} \quad \uparrow$$

$$0.2378 \times 2 = 0.4756$$

$$0.4756 \times 2 = 0.9512$$

$$0.9512 \times 2 = 1.9024$$

$$0.9024 \times 2 = 1.8048$$

$\therefore (5.2378)_{10} = (101.0011)_2$ Ans ✓

b) $(626.2555)_{10} = (626 + 0.2555)_{10}$

Now,

$$\begin{array}{r} 2 \overline{) 626} \\ 2 \overline{) 313} \quad 0 \\ 2 \overline{) 156} \quad 1 \\ 2 \overline{) 78} \quad 0 \\ 2 \overline{) 39} \quad 0 \\ 2 \overline{) 19} \quad 1 \\ 2 \overline{) 9} \quad 1 \\ 2 \overline{) 4} \quad 1 \\ 2 \overline{) 2} \quad 0 \\ 2 \overline{) 1} \quad 0 \\ \hline 0 \quad 1 \end{array} \quad \uparrow$$

$$0.2555 \times 2 = 0.511$$

$$0.511 \times 2 = 1.022$$

$$0.022 \times 2 = 0.044$$

$$0.044 \times 2 = 0.088$$

$$0.088 \times 2 = 0.176$$

$\therefore (626.2555)_{10} = (1001110010.0100)_2$ ✓

Answer to the Question NO:02

a) $(45210)_2 = (4 \times 7^4 + 5 \times 7^3 + 2 \times 7^2 + 1 \times 7^1 + 0 \times 7^0)_{10}$
 $= (11424)_{10}$

Now,

$$\begin{array}{r} 5 \overline{) 11424} \\ 5 \overline{) 2284} \quad 4 \\ 5 \overline{) 456} \quad 4 \\ 5 \overline{) 91} \quad 1 \\ 5 \overline{) 18} \quad 1 \\ 5 \overline{) 3} \quad 3 \\ \hline 0 \quad 3 \end{array} \quad \uparrow$$

$\therefore (45210)_7 = (11424)_{10} = (331144)_5$ Ans ✓

$$1) (647.284)_9 = (6 \times 9^2 + 4 \times 9^1 + 7 \times 9^0 + 2 \times 9^{-1} + 8 \times 9^{-2})_{10} \\ = (529.3264746)_{10}$$

$$\text{Now, } 5 \overline{) 529} \begin{array}{r} 105 \ 4 \\ 5 \overline{) 21 \ 0} \\ 4 \ 2 \\ 0 \ 4 \end{array}$$

$$\begin{aligned} 0.3264746 \times 5 &= 1.632373 \\ 0.632373 \times 5 &= 3.161865 \\ 0.809325 \times 5 &= 4.046625 \end{aligned}$$

$$\therefore (647.284)_9 = (529.3264746)_{10} = (4104.1304)_5$$

Answer to the Question NO: 3

$$a) (101.01010111)_2 = (0101.0101 \ 0111)_2 \\ = (5.57)_{16} \quad \text{Ans}$$

$$b) (1101000.111010)_2 = (0110 \ 1000.1110 \ 1000)_2 \\ = (68.E8)_{16} \quad \text{Ans}$$

$$c) (1.101111101011)_2 = (0001.1011 \ 1110 \ 1011)_2 \\ = (1.BEB)_{16} \quad \text{Ans}$$

Answer to the Question NO: 04

$$\begin{array}{r} 14) 2310(146 \\ \underline{14} \\ 71 \\ \underline{60} \\ 110 \\ \underline{110} \\ 0 \end{array}$$

$$\begin{array}{ccccc} 14 & 14 & 14 & 14 & 14 \\ \times 1 & \times 2 & \times 3 & \times 4 & \times 5 \\ \hline 14 & 36 & 44 & 60 & 74 \end{array}$$

$$\therefore (2310)_8 / (14)_8 = (146)_8 \\ = (1 \times 8^2 + 4 \times 8^1 + 6 \times 8^0)_{10} \\ = (102)_{10} \quad \text{(verified)}$$

$$\begin{aligned} (2310)_8 &= (2 \times 8^3 + 3 \times 8^2 + 1 \times 8^1 + 0 \times 8^0)_{10} \\ &= (1224)_{10} \\ (14)_8 &= (1 \times 8^1 + 4 \times 8^0)_{10} \\ &= (12)_{10} \\ \therefore (2310)_8 / (14)_8 &= (102)_{10} \end{aligned}$$

$$\begin{array}{r} b) (23)_7 \\ \times (12)_7 \\ \hline 46 \\ 230 \\ \hline 306 \end{array}$$

$$\begin{aligned} (23)_7 &= (2 \times 7^1 + 3 \times 7^0)_{10} \\ &= (17)_{10} \\ (12)_7 &= (1 \times 7^1 + 2 \times 7^0)_{10} \\ &= (9)_{10} \\ (17)_{10} \times (9)_{10} &= (153)_{10} \end{aligned}$$

$$\therefore (23)_7 \times (12)_7 = (306)_7 = (3 \times 7^2 + 0 + 6 \times 7^0)_{10} = (153)_{10} \text{ (verified)}$$

$$c) (BAAB)_{16} - (ACCA)_{16}$$

$$\begin{array}{r} (BAAB)_{16} \\ - (ACCA)_{16} \\ \hline (DE1)_{16} \\ = (13 \times 16^2 + 14 \times 16^1 + 1 \times 16^0)_{10} \\ = (3553)_{10} \end{array}$$

$$\begin{aligned} (BAAB)_{16} &= (11 \times 16^3 + 10 \times 16^2 + 10 \times 16^1 + 11 \times 16^0)_{10} \\ &= (47787)_{10} \\ (ACCA)_{16} &= (10 \times 16^3 + 12 \times 16^2 + 12 \times 16^1 + 10 \times 16^0)_{10} \\ &= (44234)_{10} \\ (47787 - 44234)_{10} &= (3553)_{10} \end{aligned}$$

(verified) ✓

$$d) (788)_9$$

$$\begin{array}{r} + (233)_9 \\ \hline (1132)_9 \end{array}$$

$$\begin{aligned} &= (1 \times 9^3 + 1 \times 9^2 + 3 \times 9^1 + 2 \times 9^0)_{10} \\ &= (839)_{10} \end{aligned}$$

$$\begin{aligned} (788)_9 &= (7 \times 9^2 + 8 \times 9^1 + 8 \times 9^0)_{10} \\ &= (647)_{10} \\ (233)_9 &= (2 \times 9^2 + 3 \times 9^1 + 3 \times 9^0)_{10} \\ &= (192)_{10} \\ (647 + 192)_{10} &= (839)_{10} \end{aligned}$$

(verified)

Answer to the Question No: 5

$$a) (48A)_{13} = (4 \times 13^2 + 8 \times 13^1 + 10 \times 13^0)_{10} = (790)_{10}$$

Now,

$$\begin{array}{r} 6 \overline{) 790} \\ \underline{6} \\ 191 \\ \underline{6} \\ 215 \\ \underline{6} \\ 33 \\ \underline{6} \\ 03 \end{array}$$

$$\therefore (48A)_{13} = (790)_{10} = (3359)_6$$

$$b) (10110111)_7 = (1 \times 7^7 + 0 + 1 \times 7^5 + 1 \times 7^4 + 0 + 7^2 \times 1 + 1 \times 7^1 + 1)_{10}$$

$$= (842808)_{10}$$

Now,

$$\begin{array}{r} 4 \overline{) 842808} \\ 4 \overline{) 210702 \ 0} \\ 4 \overline{) 52675 \ 2} \\ 4 \overline{) 13168 \ 2} \\ 4 \overline{) 3292 \ 0} \\ 4 \overline{) 823 \ 0} \\ 4 \overline{) 205 \ 3} \\ 4 \overline{) 51 \ 2} \\ 4 \overline{) 12 \ 3} \\ 4 \overline{) 3 \ 0} \\ \hline 0 \ 3 \end{array}$$

$$\therefore (10110111)_7 = (842808)_{10} = (3031300320)_4 \text{ Ans}$$

$$c) (0011)_{BCD} = (3)_{10}$$

$$\text{Now, } 4 \overline{) 3} \quad 3 \uparrow$$

$$\therefore (0011)_{BCD} = (3)_{10} = (3)_4 \text{ Ans}$$

$$d) (10011)_{10} = (0001 \ 0000 \ 0000 \ 0001 \ 0001)_{BCD}$$

$$= (0100 \ 0011 \ 0011 \ 0100 \ 0100)_{\text{Excess-3}}$$

Ans

Binary	Decimal
0000	0
0001	1
0011	3
0100	4

e)

Binary	Decimal
0000	0
0001	1
1000	8
0111	7

$$\begin{aligned} 2) (1100011)_{10} &= (0001 \ 0001 \ 0000 \ 0000 \ 0000 \ 0001000)_{\text{Ex}} \\ &= (1000 \ 1000 \ 0111 \ 0111 \ 0111 \ 1000 \ 1000)_{\text{Ex}} \end{aligned}$$

Answer to the Question No: 06

$$(645)_7 = (6 \times 7^2 + 4 \times 7^1 + 5 \times 7^0)_{10} \quad \left| \begin{array}{l} (327)_{10} + (293)_{10} = (620)_{10} \\ (327)_{10} - (293)_{10} = (34)_{10} \end{array} \right.$$

$$(566)_7 = (5 \times 7^2 + 6 \times 7^1 + 6 \times 7^0)_{10} \quad \left| \begin{array}{l} (327)_{10} * (293)_{10} = (95811)_{10} \end{array} \right.$$

Addition:

$$\begin{array}{r} (645)_7 \\ + (566)_7 \\ \hline (1544)_7 \end{array}$$

$$\begin{aligned} \therefore (645)_7 + (566)_7 &= (1544)_7 \\ &= (1 \times 7^3 + 5 \times 7^2 + 4 \times 7^1 \\ &\quad + 4 \times 7^0)_{10} \\ &= (620)_{10} \end{aligned}$$

(verified) ✓

subtraction:

$$\begin{array}{r} (645)_7 \\ - (566)_7 \\ \hline (46)_7 \end{array}$$

$$\begin{aligned} \therefore (645)_7 - (566)_7 &= (46)_7 \\ &= (4 \times 7^1 + 6 \times 7^0)_{10} \\ &= (34)_{10} \end{aligned}$$

(verified) ✓

Multiplication:

$$\begin{array}{r} (645)_7 \\ (566)_7 \\ \hline 5502 \\ 55020 \\ 452400 \\ \hline (546222)_7 \end{array}$$

$$\begin{aligned} \therefore (645)_7 \times (566)_7 &= (546222)_7 \\ &= (5 \times 7^5 + 4 \times 7^4 + 6 \times 7^3 + 2 \times 7^2 \\ &\quad + 2 \times 7^1 + 2 \times 7^0)_{70} \\ &= (95811)_{10} \\ &\text{(verified)} \end{aligned}$$

Answer to the Question NO: 7

$$\begin{aligned} (623)_9 &= (6 \times 9^2 + 2 \times 9^1 + 3 \times 9^0)_{10} \\ &= (507)_{10} \end{aligned}$$

$$\begin{aligned} (178)_9 &= (1 \times 9^2 + 7 \times 9^1 + 8 \times 9^0)_{10} \\ &= (152)_{10} \end{aligned}$$

$$\begin{aligned} (507)_{10} + (152)_{10} \\ &= (659)_{10} \end{aligned}$$

$$\begin{aligned} (507)_{10} - (152)_{10} \\ &= (355)_{10} \end{aligned}$$

$$\begin{aligned} (507)_{10} \times (152)_{10} \\ &= (77064)_{10} \end{aligned}$$

Addition:

$$\begin{array}{r} (623)_9 \\ + (178)_9 \\ \hline (812)_9 \end{array}$$

$$\begin{aligned} \therefore (623)_9 + (178)_9 &= (812)_9 = (8 \times 9^2 + 1 \times 9^1 + 2 \times 9^0)_{10} \\ &= (659)_{10} \end{aligned}$$

subtraction:

$$\begin{array}{r} (623)_9 \\ -(178)_9 \\ \hline (434)_9 \end{array}$$

$$\begin{aligned} \therefore (623)_9 - (178)_9 &= (434)_9 \\ &= (4 \times 9^2 + 3 \times 9^1 + 4 \times 9^0)_{10} \\ &= (355)_{10} \end{aligned}$$

verified ✓

multiplication:

$$\begin{array}{r} (623)_9 \\ \times (178)_9 \\ \hline 5506 \\ 47730 \\ 62300 \\ \hline (126636)_9 \end{array}$$

$$\begin{aligned} \therefore (623)_9 \times (178)_9 &= (126636)_9 \\ &= (1 \times 9^5 + 2 \times 9^4 + 6 \times 9^3 \\ &\quad + 6 \times 9^2 + 3 \times 9^1 \\ &\quad + 6 \times 9^0)_{10} \\ &= (77064)_{10} \end{aligned}$$

(verified)

Answer to the Question No: 8

a) $(0100101011111)_2$

$= (0100101011111)_2$

[since the number is positive and 2's can also be represent as binary number system]

$$\begin{aligned} &= (0 + 1 \times 2^{11} + 0 + 0 + 1 \times 2^8 + 0 + 1 \times 2^6 + 0 + 1 \times 2^4 + 1 \times 2^3 \\ &\quad + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0)_{10} \\ &= (2399)_{10} \checkmark \text{ Ans} \end{aligned}$$

Answer to the Question No: 9

$$22 - 13 = 22 + (-13)$$

Now,

22	13
binary unsigned: 10110	1101
+ve: 0010110	0001101
1s: 0010110	1s: 1110010
	+ 1
	(1110011) _{2s}

$$\begin{array}{r} 2 \overline{) 22} \\ \underline{2 11} \\ 2 \underline{5} 1 \\ 2 \underline{2} 1 \\ 2 \underline{1} 0 \\ \underline{0} 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 13} \\ \underline{2 6} \\ 2 \underline{3} \\ 2 \underline{1} 1 \\ 0 1 \end{array}$$

$$\begin{array}{r} (22)_{10} \Rightarrow (0010110)_{2s} \\ - (13)_{10} \Rightarrow \phantom{(0010110)_{2s}} + (1110011)_{2s} \\ \hline (10001001)_{2s} \end{array}$$

Since the carry bit 1 is discarded,

$$\text{So, } 22 + (-13) = (0001001)_{2s} \text{ Ans}$$

Here, we added two different sign numbers, so there will be no overflow. ✓

Answer to the Question NO: 10

$$(98) - (45) = (98) + (-45)$$

$$\begin{array}{r} 2 \overline{) 98} \\ 2 \overline{) 49} \ 0 \\ 2 \overline{) 24} \ 1 \\ 2 \overline{) 12} \ 0 \\ 2 \overline{) 6} \ 0 \\ 2 \overline{) 3} \ 0 \\ 2 \overline{) 1} \ 1 \\ 0 \ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 45} \\ 2 \overline{) 22} \ 1 \\ 2 \overline{) 11} \ 0 \\ 2 \overline{) 5} \ 1 \\ 2 \overline{) 2} \ 1 \\ 2 \overline{) 1} \ 0 \\ 0 \ 1 \end{array}$$

98	45
Binary signed : 1100010	101101
+ve: 0000011001	000000101101
2's: 000001100010	(111111010010) _{1s}
	+ 1
	(111111010011) _{2s}

$$\begin{array}{r} 98 \\ -45 \\ \hline 000001100010 \\ (+) 111111010011 \\ \hline 1000000110101 \end{array} \text{ Ans}$$

Since we ~~are~~ discard carry bit, and adding two different signed numbers, there will be no overflow.

Answer to the Question NO: 11

$$(19 + 47)$$

$$\begin{array}{r} 2 \overline{) 19} \\ 2 \overline{) 9} \ 1 \\ 2 \overline{) 4} \ 1 \\ 2 \overline{) 2} \ 0 \\ 2 \overline{) 1} \ 0 \\ 0 \ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 47} \\ 2 \overline{) 23} \ 1 \\ 2 \overline{) 11} \ 1 \\ 2 \overline{) 5} \ 1 \\ 2 \overline{) 2} \ 1 \\ 2 \overline{) 1} \ 0 \\ 0 \ 1 \end{array}$$

19	47
Binary signed : 10011	101111
+ve: 00010011	00101111

$$\begin{array}{r} 47 \\ (+) 19 \Rightarrow \\ \hline 00010011 \\ + 00101111 \\ \hline 01000010 \end{array}$$

we added two positive numbers and the result is also positive, so, there is no overflow.

$$\therefore (19) + (47) = (01000010)_{25} \text{ Ansr}$$

Answer to the Question No:12

$$91 - 499 = 91 + (-499)$$

$$\begin{array}{r} 2 \overline{) 91} \\ 2 \overline{) 451} \\ 2 \overline{) 221} \\ 2 \overline{) 110} \\ 2 \overline{) 51} \\ 2 \overline{) 21} \\ 2 \overline{) 10} \\ 01 \end{array}$$

$$\begin{array}{r} 2 \overline{) 499} \\ 2 \overline{) 2491} \\ 2 \overline{) 1241} \\ 2 \overline{) 620} \\ 2 \overline{) 310} \\ 2 \overline{) 151} \\ 2 \overline{) 71} \\ 2 \overline{) 31} \\ 2 \overline{) 11} \\ 01 \end{array}$$

91	499
1011011	11111011
25: 000001011011	000011111011
	15: 1111000001100
	+ 1
	(1111000001101) ₂₅

$$\begin{array}{r} 25: \\ (000001011011)_{25} \\ (1111000001101)_{25} \\ \hline (1111001101000)_{25} \checkmark \end{array}$$

$$\begin{array}{r} 15: \\ (000001011011)_{25} \\ (1111000001100)_{25} \\ \hline (1111001100111)_{25} \end{array}$$

Here,
adding two different signed numbers,
so there will be no overflow.

$$b) (379) + (98)$$

$$\begin{array}{r}
 2 \overline{) 379} \\
 \underline{2 \overline{) 189} \ 1} \\
 \underline{2 \overline{) 94} \ 1} \\
 \underline{2 \overline{) 47} \ 0} \\
 \underline{2 \overline{) 23} \ 1} \\
 \underline{2 \overline{) 11} \ 1} \\
 \underline{2 \overline{) 5} \ 1} \\
 \underline{2 \overline{) 2} \ 1} \\
 \underline{2 \overline{) 1} \ 0} \\
 0 \ 1
 \end{array}$$

$$\begin{array}{r}
 2 \overline{) 98} \\
 \underline{2 \overline{) 49} \ 0} \\
 \underline{2 \overline{) 24} \ 1} \\
 \underline{2 \overline{) 12} \ 0} \\
 \underline{2 \overline{) 6} \ 0} \\
 \underline{2 \overline{) 3} \ 0} \\
 \underline{2 \overline{) 1} \ 1} \\
 0 \ 1
 \end{array}$$

$$\begin{array}{r|l}
 379 & 98 \\
 \hline
 101111011 & 1100010 \\
 + 000010011011 & 00000110010 \\
 \hline
 \end{array}$$

Since both numbers are positive there 1's and 2's complement are same as the signed numbers

$$\begin{array}{r}
 379 \\
 + 98 \\
 \hline
 \Rightarrow \begin{array}{r}
 000010111011 \\
 (+) 0000001100010 \\
 \hline
 0000111011101
 \end{array}
 \end{array}$$

Since two same signed numbers and after adding the sign is unchanged so no overflow.